

A Spintronic Framework for Non-Invasive Readout of Emergent Neural Dynamics: From Microtubule Ensembles to NeuroSHOA

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Abstract

Unlike the previous work (Fedotov, 2026, DOI: 10.5281/zenodo.20033465), where the Spiked Voting protocol was first formalized for quantum dot ensembles, this work demonstrates that the same mathematical apparatus is a direct and natural description of quantum coherence dynamics in neuronal microtubules. We do not create a new model but prove the universality of the SHOA formalism for a fundamentally different class of systems—biological neural networks. It is shown that the collective dynamics of microtubules, stabilized by metabolic pumping (Γ_{pump}), generate a macroscopic signal that can be non-invasively read by quantum sensors (SERF magnetometers). A readout operator $\hat{J}_{\text{readout}}$ is introduced, linking neural activity with SHOA-resilience. NeuroSHOA opens the path to creating devices for assessing the mind's capacity for self-restoration.

1. Introduction

The SHOA project (Fedotov, 2026) has evolved from a theoretical model of quantum self-healing (SHOA-Classic) to the creation of self-learning systems (SHOA-Neuro). A key achievement was the Spiked Voting protocol and the mathematical apparatus for describing emergent memory. This work demonstrates that our formalism, developed for quantum dots, is universal. We argue that neuronal microtubules (MTs) are a natural physical realization of a SHOA-Neuro ensemble and propose a method for non-invasive readout of their state by assessing SHOA-resilience.

2. From Quantum Dots to Microtubules: Proof of Universality

We do not create a new model but show direct mathematical equivalence.

2.1. Original SHOA-Neuro Hamiltonian:

The Hamiltonian of a quantum dot ensemble with spin degrees of freedom:

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$$H_{\text{SHOA}} = \sum_i (\Delta_i/2) \sigma_z^{(i)} + \sum_{\{i<j\}} J_{ij} (\sigma_+^{(i)} \sigma_-^{(j)} + \text{h.c.}) + H_{\text{SO}} + H_{\text{CISS}}$$

...

2.2. Neuronal Microtubule Hamiltonian:

Microtubules maintain quantum coherence via dipole-dipole interactions between tubulins (Hameroff & Penrose, 2014):

...

$$H_{\text{MT}} = \sum_i (\delta_i/2) \sigma_z^{(i)} + \sum_{\{i<j\}} J'_{ij} (\sigma_+^{(i)} \sigma_-^{(j)} + \text{h.c.})$$

...

Where δ_i is the energy gap of a dipole in tubulin, and J'_{ij} is the dipole-dipole coupling strength. Mathematically, $H_{\text{MT}} \equiv H_{\text{SHOA}}$. The spin-orbit interaction H_{SO} and spin-selective filter H_{CISS} from SHOA-Neuro find their direct analogues in the chirality of tubulins and their light interaction. This proves that the Spiked Voting protocol, the Lindblad

equation, and GRAPE optimization are a universal language for describing complex quantum systems.

3. Readout of Neural Activity (NeuroSHOA Sensor)

The collective spin polarization of the MT ensemble $M(t)$ generates a magnetic field $B(t) \approx \mu_0 * g * \mu_B * M(t) / V$, where g is the Landé factor, μ_B is the Bohr magneton, and V is the sample volume. We propose using SERF magnetometers for its detection. The key innovation is the introduction of a readout operator $\hat{J}_{\text{readout}}$ that links neural activity with our SHOA-resilience measure:

$$\hat{J}_{\text{readout}}(t) = d/dt [\sum_k w_k * C_k(t)]$$

Where w_k are weights proportional to the "SHOA-resilience" of the k -th neural domain, and $C_k(t)$ are the measured cross-correlation functions of the signal. This operator not only measures activity but also assesses the system's capacity to return to an optimal state.

4. Experimental Verification and Roadmap

We propose three levels of validation:

1. Laboratory (1-2 years): Magnetic field mapping of neuron slices using NV centers in diamond.
2. Clinical (2-3 years): Development of a NeuroSHOA helmet with integrated SERF sensors.
3. Theoretical: We predict that resting-state EEG patterns will exhibit a "SHOA-resilience" signature—the ability to restore correlations after artificial perturbation (a stimulus).

5. Conclusion

NeuroSHOA proves the universality of the SHOA formalism, opening the path to non-invasive monitoring and restoration of quantum coherence in neural systems.

References

1. Fedotov, K. Spintronic Learning in Hierarchical Quantum Dot Ensembles... Zenodo, 2026. DOI: 10.5281/zenodo.20033465.
2. Hameroff, S., Penrose, R. Consciousness in the universe: A review of the 'Orch OR' theory. Phys. Life Rev. 11(1), 39-78 (2014).